How accurate is altitude from pressure geolocators? A global assessment of three pressure-to-altitude conversions on the ground and in flight

Raphaël Nussbaumer[1*] and Mathieu Gravey[2]

[1*]Swiss Ornithological Institute, Seerose 1, Sempach, 6204, Switzerland.

[2]Institute for Interdisciplinary Mountain Research, Austrian Academy of Sciences, Innsbruck, Austria.

*Corresponding author(s). E-mail(s): raphael.nussbaumer@vogelwarte.ch;

Contributing authors: mathieu@mgravey.com;

Abstract

Background: Pressure geolocators record atmospheric pressure throughout the year on small birds, and pressure sensors also provide vertical records in GPS-equipped tags. The altitudes derived from these records reveal migration strategies and inform conservation assessments, including exposure to wind turbines. However, pressure-to-altitude conversions make assumptions about atmospheric conditions whose consequences for altitude error remain poorly quantified.

Methods: We evaluated three conversions: a fixed standard atmosphere, a conversion using local ERA5 surface conditions, and an ERA5 conversion additionally accounting for virtual temperature and a climatological temperature lapse rate from GPT3. Treating reference pressures as exact conversion inputs, we compared estimates with 1,983 GHCNh weather stations worldwide and IGRA2 radiosondes from 648 stations up to 6 km above ground. We summarize absolute errors by their medians and 95th percentiles, distinguish station bias from within-station SD, and weight flight-height errors by the flight time of 203 tracked birds.

Results: At screened ground-reference stations, the standard atmosphere produced a median absolute error of 41.0 m, with 5% of absolute errors exceeding

143 m. Local ERA5 conditions reduced these to 3.9 m and 22 m, respectively. In flight, flight-time-weighted median absolute errors were 21.8 m with the standard atmosphere, 7.4 m with ERA5, and 5.9 m after adding virtual temperature and GPT3. At 2–3 km, the correction reduced the median station bias from -32.2 m to -4.5 m. At 2–3 km above ground, the 95th percentile of absolute flight-height error fell from 238 m with the standard atmosphere to 86 m with the corrected ERA5 conversion.

Conclusions: Local atmospheric conditions substantially reduce the large altitude errors introduced by a fixed standard atmosphere; virtual temperature and a climatological lapse rate further reduce systematic flight-height errors. The conversion is computationally simple and available in a software function using measured pressure, measurement time and an estimated geographic position. Remaining atmospheric and sensor errors must still be considered when interpreting flight behaviour or classifying flights relative to turbine heights.

Keywords: biologging, barometric altimetry, flight altitude, ERA5, radiosonde, multi-sensor geolocator, GeoPressureR

1 Background

Altitude is a central dimension of how birds use their environment. In flight, it helps explain how migrants respond to winds and temperature [1, 2], how birds partition aerial foraging space [3], and how they use soaring flight [4, 5]. Vertical records also reveal behaviours whose functions remain incompletely understood, including twilight ascents in swifts [6], exploratory climbs during migration [7], and pronounced day–night changes in flight altitude [8, 9]. On the ground, elevation describes habitat use and movements such as elevational commuting, uphill roosting and seasonal migration [10–12]. Height distributions also inform conservation assessments, including exposure to wind turbines [13, 14].

Miniature pressure sensors make it possible to study these vertical movements over long deployments, including in small migratory birds carrying multi-sensor geolocators [15–17]. Pressure sensors are also incorporated into GPS-equipped tags, extending their applications to larger birds [4, 5, 13]. Alongside these biological applications, pressure-derived elevation can help constrain geographic positions during track reconstruction [18]. Interpreting these records requires knowing how accurately pressure can be converted into altitude, and whether the resulting uncertainty is small enough to resolve the movements or height differences of interest [19].

Pressure-to-altitude conversion requires a reference pressure at a known altitude and an estimate of the temperature profile between that level and the bird. The simplest approach uses a fixed standard atmosphere [20], as in several geolocator studies [7, 17, 21, 22]. Other studies obtain reference conditions from local weather stations [3], reanalysis sea-level pressure [6], or periods when birds are assumed to be at the sea

surface [13, 23]. Reanalysis can also supply the temperature information needed for conversion: GeoPressureR uses ERA5 surface pressure, temperature and orography at an estimated position [18, 24, 25], while Schaub et al. [26] additionally estimated the temperature lapse rate from surface and upper-air conditions.

Previous evaluations have established the performance of pressure-derived heights in controlled device trials [27], at grounded birds [26], and through comparisons with GPS during flight [23]. These studies provide important evidence about particular devices, conversion methods and field conditions. However, their results do not establish how atmospheric conversion errors vary across the climates, terrain and flight heights encountered during long tracking deployments. Quantifying this component separately helps explain when errors arise from the atmospheric assumptions and how much changing those assumptions can improve altitude estimates.

Here, we compare three pressure-to-altitude conversions: a fixed standard atmosphere, an ERA5 surface reference with the standard lapse rate, and the same reference with virtual temperature and a climatological lapse rate from GPT3 [28]. Using worldwide weather stations, stations with documented pressure-sensor heights and radiosonde profiles up to 6 km above the ground, we quantify atmospheric conversion errors on the ground and in flight. We distinguish systematic bias from temporal variability, characterize the range of errors, and summarize flight-height errors using the heights of tracked birds. Our aim is to establish how much local atmospheric information and a more realistic temperature profile improve a conversion suitable for long pressure records.

2 Methods

2.1 Three conversions from pressure to altitude

Pressure decreases with height because each layer of air supports the weight of the air above it. Combining this hydrostatic approximation with the ideal gas law gives the hypsometric relation [29]:

$$z = z_0 + \frac{R\,\bar{T}_v}{g\,M}\,\ln\frac{p_0}{p}, \tag{1}$$

with z the altitude (m) at pressure p, z_0 and p_0 the altitude and pressure of a reference level, $g = 9.80665$ m s^{-2}, $R = 8.31432$ J mol^{-1} K^{-1}, $M = 0.0289644$ kg mol^{-1}, and $\bar{T}_v$ the mean virtual temperature.

The virtual temperature $T_v = T(1+0.608\,q)$, with T the air temperature in kelvin and q the specific humidity, accounts for moist air being lighter than dry air. Temperature can vary with height; the conversion uses the mean virtual temperature between the reference level and the bird, weighted by increments of log pressure:

$$\bar{T}_v = \frac{1}{\ln(p_0/p)}\int_{p}^{p_0} T_v(p')\,d\ln p'.$$

To simplify the computation, temperature is assumed to change at a constant rate with height, giving a linear profile $T(z) = T_0 + L(z - z_0)$, where T_0 is the reference temperature in kelvin. This rate, called the lapse rate $L = dT/dz$ (K m^{-1}), is negative when temperature decreases with height. Without a humidity correction, $T_v = T$, and integration gives the barometric equation:

$$z = z_0 + \frac{T_0}{L}\left[\left(\frac{p}{p_0}\right)^{-RL/(gM)} - 1\right]. \tag{2}$$

The three conversions differ in their reference level and in the column temperature they assume (Figure 1):

1. **Standard atmosphere.** The reference is sea level ($z_0 = 0$, $p_0 = 1013.25$ hPa) with $T_0 = 288.15$ K (15 °C) and $L = -6.5$ K km^{-1} [20]. Its main advantage is that it needs no external data, but these fixed conditions do not account for changes in pressure, temperature or humidity.
2. **ERA5, standard lapse rate.** The ERA5 reanalysis provides surface pressure p_0, 2 m air temperature T_0 and surface geopotential Φ at the location and hour of each measurement, with reference altitude $z_0 = \Phi/g$. This accounts for local pressure and temperature variations, while retaining a constant lapse rate of -6.5 K km^{-1} and no humidity correction.
3. **ERA5, virtual temperature and GPT3 lapse rate.** This conversion adds two refinements to the ERA5 reference. First, humidity at the reference level is accounted for by replacing T_0 in Equation 2 with the reference-level value of T_v. The specific humidity q needed for this correction is calculated from ERA5 surface pressure p_0 and 2 m dewpoint T_d, via vapour pressure e [30]:

$$q = \frac{0.622\,e}{p_0 - 0.378\,e}, \qquad e = 611.2 \exp\left(\frac{17.67\,T_d}{T_d + 243.5}\right),$$

where T_d is in °C and p_0 and e are in Pa. Second, the standard lapse rate is replaced with a location- and season-dependent value from GPT3, an empirical troposphere model that provides a lapse rate for any location and time of year [28]. Values are interpolated bilinearly on the 1° grid. This formulation does not resolve humidity aloft.

All three conversions are implemented in R. Both ERA5 conversions are available in the GeoPressureR function `pressure_to_altitude()` (`altitude_formula = "standard"` or `"virtual"`). ERA5 single-level variables [31] are read from the ECMWF ARCO archive at the nearest 0.25° grid cell and hour.

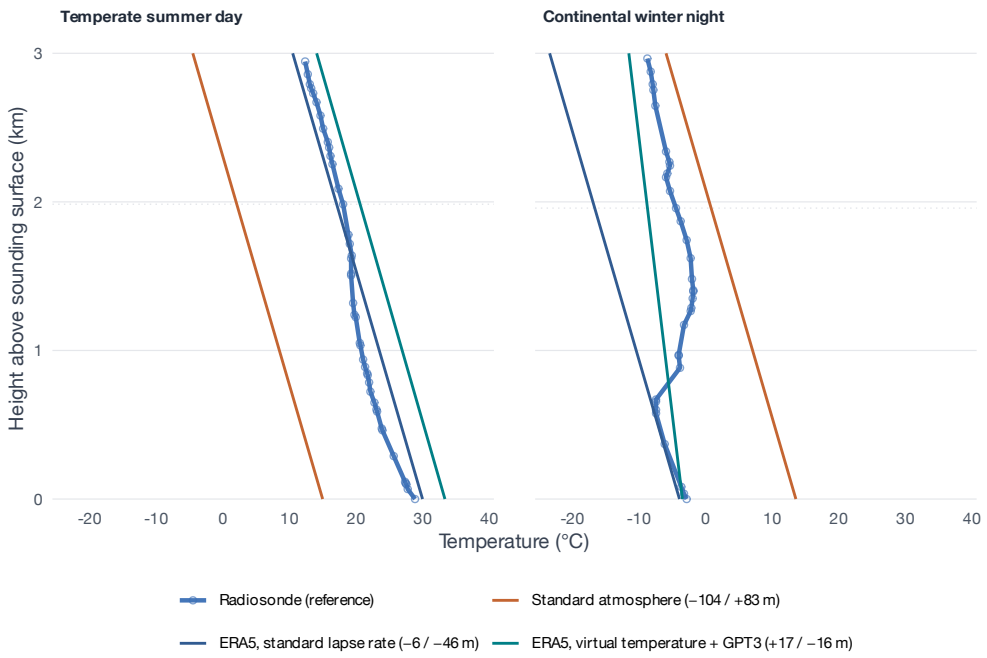


Figure 1: Radiosonde temperatures and the temperature profiles assumed by the three pressure-to-altitude conversions for a temperate summer day (left) and a continental winter night (right). Colours identify each profile; the legend also gives its signed height error near 2 km above the sounding surface (summer / winter), after subtracting the surface error. The radiosonde is the reference profile. The examples were selected because their errors were close to the class medians.

2.2 Reference datasets

We test pressure-to-altitude conversion in two settings: at weather stations with reported elevations, and during balloon ascents, where measured pressure can be compared with the reported height profile. Figure 2 shows the locations used in both comparisons; full data-screening criteria are in Additional file 1: S1.

2.2.1 Ground-level reference: GHCNh

For the ground comparison, we use NOAA's Global Historical Climatology Network hourly (GHCNh) archive [32], which provides hourly surface pressure and station elevation. We use each station's most recent sufficiently complete year from 2022–2026, with stations distributed globally and additional representation of high elevations. To ensure the reported elevation is a useful reference, we remove pressure observations outside 300–1100 hPa or marked as suspect by GHCNh, then exclude stations with

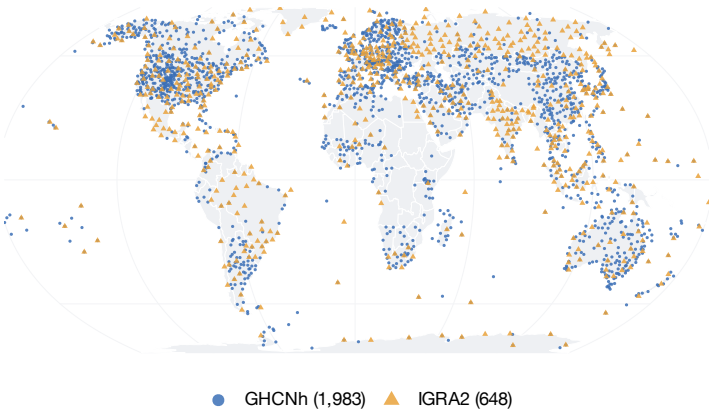


Figure 2: Weather stations and radiosonde launch sites used to evaluate pressure-derived altitude at the surface and through the atmosphere. Station counts are shown in brackets; complementary ground references are described in Additional file 1: S4.

large persistent pressure-derived elevation offsets from ERA5, abrupt changes or frequent implausible readings. This retains 1,983 stations; full thresholds are given in Additional file 1: S1. Each retained pressure report is paired with ERA5 conditions from the nearest hour.

Because listed station elevation may not exactly match pressure-sensor height, we also assess two ground references with more explicit height information: pressure sensors at GNSS stations with checked site-log metadata and MeteoSwiss stations with published pressure-sensor heights. These comparisons test how reference-height information affects the results and are presented in Additional file 1: S4.

2.2.2 Flight-height reference: IGRA2

To assess how conversion error changes above the surface, we use balloon soundings from NOAA's Integrated Global Radiosonde Archive, version 2 (IGRA2) [33–35]. Radiosondes measure pressure as they rise and report the corresponding atmospheric height. We compare pressure-derived height changes with reported height changes, from the surface to 6 km above it, using 648 stations observed in 2023–2024. Each profile level is paired with ERA5 conditions from the nearest hour.

2.3 Error metrics

We treat reference pressures as exact conversion inputs and define error as estimated minus reference height. We use the same summaries on the ground and in flight. For **individual observations**, the median absolute error describes a typical error magnitude; the 95th percentile of absolute error describes the upper tail. For **stations**,

bias is the mean signed error and SD describes variability around that mean. We report weighted medians across stations: absolute station bias, |bias|, on the ground, and signed station bias in flight to retain the direction of height error. Median |bias| describes a persistent station offset; it differs from the median absolute error of individual observations.

On the ground, error is converted altitude minus the reported reference elevation. Individual hourly errors describe single reports; station bias and SD separate persistent offsets from temporal variability. Each retained 2°×2° GHCNh cell receives equal weight, shared among its stations and then among each station's observations; GNSS and MeteoSwiss stations receive equal weight. All three conversions use the same records and weights. Because the GHCNh plausibility screen uses ERA5 residuals, we also report results without that screen (Additional file 1: S4). Terrain, season, local time and climate are assessed in Additional file 1: S2.

In flight, error is the converted height difference from the sounding surface minus the corresponding radiosonde height difference. Individual absolute errors are summarized within height bands. Station bias and SD are calculated across retained levels and soundings within each station–height band, separating persistent offsets from variation among profiles and levels. Figure 4 shows their weighted medians and station interquartile ranges in 250 m bands. Individual-level weights divide each station's spatial weight among all its retained levels and are then normalized within each reporting band; alternative within-band weights are assessed in Additional file 1: S4.

For the overall bird summary, we use the same median absolute error, with each height band's error distribution weighted by its share of recorded flight time. These shares are estimated from 203 tagged birds of 25 species in the Master GeoLocator Data Package [36], accounting for the sampling interval represented by each record (Figure 4 c). The weighted median is calculated from the combined individual-error distribution, rather than by averaging band medians. It describes this sample's height distribution, not an individually validated bird flight. Alternative biological weights are assessed in Additional file 1: S4.

3 Results

3.1 Ground altitude errors

Across 1,983 screened GHCNh stations, the median / 95th percentile of absolute hourly errors fell from 41.0 / 143 m with the standard atmosphere to 3.9 / 22 m with ERA5 and the standard lapse rate (Figure 3). These pooled errors describe individual reports; the station summaries separate persistent offsets from temporal variability. The weighted median station |bias| and SD fell from 22.8 and 54.1 m, respectively, to 2.9 and 3.5 m with ERA5. Adding virtual temperature and the GPT3 lapse rate produced little further improvement at ground stations.

Station offsets and variability were larger in rougher terrain: median |bias| rose from 2.3 m in flat terrain (sub-grid ERA5 orography SD below 20 m) to 7.2 m in mountains

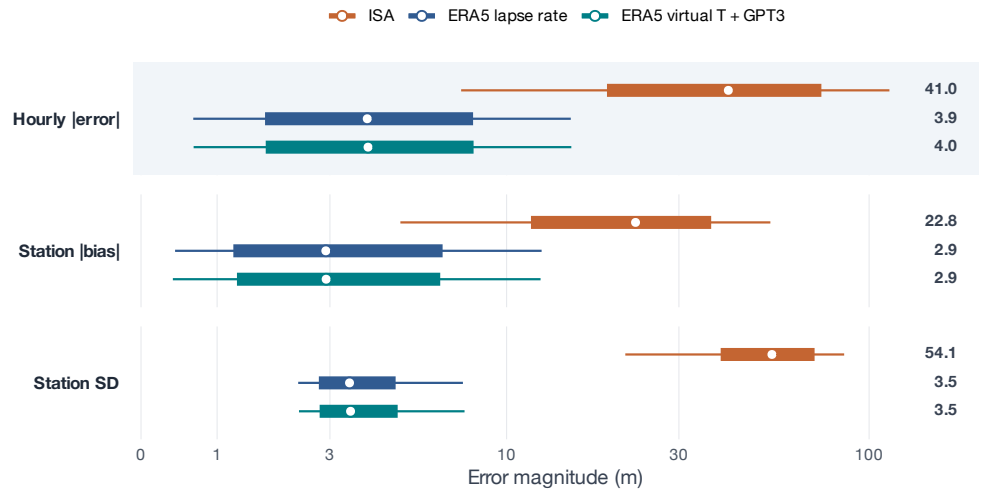


Figure 3: Ground errors at screened GHCNh stations. The lightly shaded top panel shows the weighted distribution of absolute errors across hourly observations. The lower panels group station-level absolute bias and within-station SD, separating persistent offsets from temporal variability. Thick segments show the weighted 25th–75th percentiles, thin segments the 10th–90th percentiles, and white dots mark weighted medians; numbers at the right give those medians in metres. All panels share a pseudo-log x-axis (linear near zero and logarithmic for larger errors; sigma = 1 m); each 2°×2° cell contributes equal total weight.

(above 300 m); station SD rose from 3.0 to 7.9 m. The ERA5 median absolute hourly error was similar with the references that provide more explicit pressure-sensor heights: 3.0 m at 49 GNSS stations and 4.2 m at 138 MeteoSwiss stations, compared with 3.9 m at GHCNh (Additional file 1: S4). At 24 matched Swiss stations, substituting published pressure-sensor height for listed GHCNh elevation reduced median |bias| from 6.0 to 3.0 m, close to 2.7 m with MeteoSwiss pressure. This indicates that height metadata can account for part of the remaining offset at these sites, without establishing the cause of discrepancies at other stations (Additional file 1: S4).

3.2 Flight-height errors

At individual profile levels, median absolute error at 2–3 km above the sounding surface was 98.1 m with the standard atmosphere, 33.7 m with ERA5 and the standard lapse rate, and 25.6 m with virtual temperature and GPT3. The corresponding 95th percentiles of absolute error were 238, 120 and 86 m (Additional file 1: S1). At station level, Figure 4 separates signed bias from within-station SD: the corrected conversion brings the median station bias towards zero, while variation among profiles and levels remains. At 2–3 km, adding virtual temperature and GPT3 shifted median station bias from -32.2 to -4.5 m, while median station SD remained similar (33.7 and 33.6

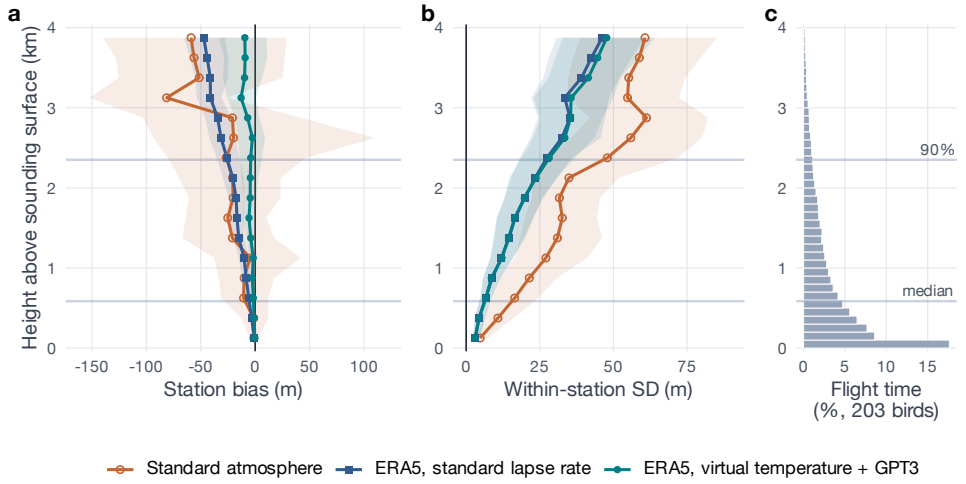


Figure 4: Surface-referenced flight-height errors against height above the sounding surface: (a) station bias (mean error; negative: height underestimated) and (b) station error SD within each height band. Lines and points: cell-weighted medians across stations in 250 m height bands; shaded bands: weighted 25th–75th percentiles for all three conversions. Curves join observed-band summaries without smoothing; the highest band is centred at 3.875 km. These station quantiles differ from the pooled individual-error summaries. The display covers 0–4 km; summaries in the text use the full evaluated range to 6 km. (c) Flight-time height distribution of 203 birds, with its median (590 m) and 90th percentile (2,350 m) marked across the panels.

m). These station summaries use the full 2–3 km reporting band, while Figure 4 uses finer bands.

After weighting profile-level errors by the tracked birds' flight-time distribution across height bands, median absolute error fell from 21.8 to 7.4 and 5.9 m. All these errors are referenced to each sounding's own surface and exclude the absolute ground-reference error.

4 Discussion

For bird tracking, our results support combining a local ERA5 surface reference with virtual temperature and a GPT3 lapse rate. The surface reference supplies the largest improvement over a fixed standard atmosphere; the temperature correction further reduces systematic flight-height bias. Using these atmospheric inputs improves the interpretation of pressure records, while sensor, position and terrain errors remain additional parts of the measurement process.

Table 1: Summary of evaluated conversion errors (m). Ground: median absolute station bias in flat terrain and mountains (sub-grid ERA5 orography SD below 20 m and above 300 m), and median within-station SD, at screened GHCNh stations. Flight: median absolute profile-level error relative to the sounding surface; the final row additionally weights the combined error distribution by recorded bird flight time. All values exclude added tag-sensor error; the ground comparison retains reference-height uncertainty.

Quantity	Standard atm.	ERA5 standard	ERA5 virtual + GPT3
Ground, flat terrain (median |bias|)	20.6	2.3	2.4
Ground, mountains (median |bias|)	35.7	7.2	7.4
Within-station scatter (median SD)	54.1	3.5	3.5
Flight, 1–1.5 km (median |error|)	54.9	14.4	10.0
Flight, 2–3 km (median |error|)	98.1	33.7	25.6
Flight-time weighted (median |error|)	21.8	7.4	5.9

4.1 Choosing and interpreting a conversion

ERA5 single-level conditions should be preferred to the fixed standard atmosphere for the applications evaluated here. On the ground, adding virtual temperature and GPT3 produces little further improvement, particularly where the station lies close to model-surface elevation and only a short column is converted. Persistent offsets and temporal scatter remain greater in rough terrain. ERA5-Land's finer grid did not improve this comparison and produced larger persistent offsets (Additional file 1: S3); spatial resolution alone is therefore an insufficient reason to choose a weather product.

For flight height, virtual temperature and GPT3 reduce the systematic underestimation remaining with the standard lapse rate. At 2–3 km, median station bias moves towards zero while median within-station SD changes little. A climatological profile can correct recurring departures from the standard lapse rate without reproducing each sounding's conditions. The seasonal and day/night profiles support this interpretation (Additional file 1: S2), extending the temperature-dependent departures noted along great-snipe migration routes [9].

Table 1 summarizes the evaluated errors for different uses. Persistent station offsets, within-station variability and individual flight-height errors answer different questions. An offset can partly cancel when comparing observations at one site, but temporal variability remains; station SD is not itself the error of a height difference. The flight-time-weighted summary gives greater influence to commonly occupied heights in the tracked sample. Its smaller value should not be applied to rare high flights, whose errors and upper tails are larger.

4.2 Interpreting bird altitude and ground clearance

The appropriate uncertainty depends on the biological comparison. Standard-atmosphere errors can change sign across climates and seasons, so a small pooled signed bias can conceal large individual errors. Comparisons among seasons, locations or times of day can inherit atmospheric differences even when the same device and conversion are used throughout. For questions about migration altitude or atmospheric-layer use, the corrected conversion reduces this source of systematic variation, but the remaining individual-error tail still matters: at 2–3 km, 5% of its absolute surface-referenced errors exceed 86 m (Additional file 1: S1).

Local ground clearance requires terrain or sea-surface elevation in a consistent vertical datum [19, 37]. Our flight comparison measures separation from a sounding's own surface; it does not validate terrain subtraction along a moving bird's track. Position uncertainty affects both weather annotation and the surface selected for subtraction. GeoPressureR's `path2elevation()` can summarize possible terrain elevations along a reconstructed path, but those summaries do not identify the ground directly beneath the bird [25]. The position-displacement pilot also shows that small changes in pooled median error can conceal larger changes for individual atmospheric queries (Additional file 1: S5). Its ten-station, near-2-km design does not establish an acceptable location-error radius.

These distinctions are especially consequential near turbine rotor boundaries. Alternative height estimates can materially change modelled collision risk [23]. At 100–250 m, the corrected conversion's surface-referenced absolute-error p95 is 22 m, but this excludes the absolute surface-reference error and added sensor error. It should not be used as the complete uncertainty of a bird's clearance above a turbine site. Rotor-zone exposure estimates require uncertainty in the full height estimate; our study did not test rotor-zone classification or collision risk directly.

4.3 Calibration and additional atmospheric information

Known-height events can complement weather-based conversion. Floating periods have been used to update pressure references in gannets and gulls [13, 23]. Such anchors can absorb part of the sensor and reference-pressure discrepancy at that event, provided behaviour and reference height are identified reliably. They do not determine the temperature profile aloft, and an additive pressure offset does not cancel exactly from altitude differences because pressure-to-height sensitivity changes with altitude. Near sea level, 1 hPa corresponds to about 8 m under typical conditions. Sensor offset, drift and relative error remain additional components; we did not compare the conversion with repeated local calibration.

Existing device and field studies address complementary parts of this problem. Lato et al. [27] used a laser reference during locally calibrated drone trials, while Schaub et al. [26] assessed barometric heights at birds inferred to be on the ground, already using ERA5 and a pressure-level-derived lapse rate. Their errors and calibration conditions differ from ours, precluding a direct accuracy ranking. Humidity and lapse-rate

corrections also have established atmospheric and GNSS precedents [28, 38]. Combining surface variables with a cached GPT3 lapse rate limits data requirements for long records. Time-specific profile methods, as used in animal tracking and aviation [26, 39], could better represent inversions; a comparison of their accuracy and retrieval cost would establish when the additional information is worthwhile.

4.4 Limits of transfer to tracking records

Ground errors are conditional on reference selection. The GHCNh screen uses ERA5 residuals and can exclude genuine model discrepancies as well as suspect pressure or height records. Removing it changes central errors less than their upper tails and preserves the improvement over the standard atmosphere (Additional file 1: S4). The paired Swiss height-only comparison supports a metadata contribution locally, without diagnosing discrepancies elsewhere. GNSS and MeteoSwiss supply more explicit height information, but do not establish independence from ERA5. Screened ground errors should therefore be read alongside the selection sensitivity.

The flight comparison likewise does not establish independent geometric-height accuracy. Reported radiosonde geopotential heights may be derived from measured atmospheric profiles, and we did not recover the height-generation method for each level. ERA5 assimilates conventional surface and upper-air observations [24], but we did not determine which evaluated records were assimilated. Balloon drift is ignored, reference coverage is uneven, and all references are on land. Generalization to poorly represented regions and offshore conditions remains to be tested.

Finally, the bird-weighted error describes the sampled flight-time height distribution, rather than all birds or a particular trajectory. Its height weights depend on conversion and terrain assumptions. Alternative biological targets and reference weights preserve the conversion ranking, while some high-altitude tails vary (Additional file 1: S4). For practical use, report the conversion and vertical reference, select error summaries appropriate to the height and biological question, and account for sensor, position and terrain uncertainty alongside atmospheric error. This applies to pressure-derived heights in both small geolocators and GPS-equipped tags; the total uncertainty will depend on the deployment.

5 Conclusions

A fixed standard atmosphere introduces substantial errors into pressure-derived bird altitudes. At screened ground-reference stations, its median absolute error was 41.0 m and 5% of absolute errors exceeded 143 m. Referencing the conversion to local ERA5 conditions reduced these to 3.9 and 22 m. In flight, the standard-atmosphere 95th percentile of absolute error reached 238 m at 2–3 km above ground, even after referencing each flight height to its own surface. ERA5 greatly reduced these errors, while virtual temperature and a GPT3 lapse rate further reduced flight-time-weighted median

absolute error from 7.4 to 5.9 m. These improvements support more reliable interpretation of migration altitude and reduce one source of uncertainty in conservation assessments.

The corrected conversion is computationally simple: it combines surface atmospheric variables with a climatological lookup and is available in a single software function. Users supply measured pressure and obtain atmospheric conditions using an estimated bird position at the measurement time; no measured temperature profile aloft is required. This makes the correction practical for long geolocator records. It substantially reduces conversion error, while day-to-day temperature profiles, unresolved terrain, pressure-sensor errors and position uncertainty still constrain individual altitude estimates, particularly near ecological or turbine-height thresholds.

6 Declarations

Ethics approval and consent to participate. Not applicable: no animals were handled for this study; the bird flight heights come from published tracking data.

Consent for publication. Not applicable.

Availability of data and materials. Code, derived tables, figures and the downloaded data needed to rerun the analysis (all but GHCNh) are archived on Zenodo [40] and at https://github.com/GeoPressure/altitude-validation, with the earlier exploratory report preserved in the repository archive. The bird weighting uses the open-access Master GeoLocator Data Package, version 2026.0 [36]; the release-specific DOI and verified input checksums accompany the derived weights. The conversions are implemented in GeoPressureR [41]. GHCNh, IGRA2, GNSS, MeteoSwiss and ERA5 data are openly available from their providers (Additional file 1: S1).

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. RN designed the study, performed the analysis and wrote the manuscript. MG contributed to the implementation of the conversions in GeoPressureR and GeoPressureAPI and revised the manuscript. [To be confirmed.]

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